

SOFTWARE QUALITY ANALYSIS AND BUG REPORT

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Application Under Test: SauceDemo - saucedemo.com

Testing Environment: Google Chrome on Windows 11

Date: March 2026

1. EXECUTIVE SUMMARY

The objective of this report is to provide a comprehensive quality analysis of the SauceDemo e-commerce application. By following the Software Testing Life Cycle, I have executed multiple test cases using all available user profiles. This analysis focuses on Functional Testing, UI/UX Consistency, Performance, and Security/Access Control. While the system is stable for standard users, several critical defects were identified in other profiles, which are documented in this report.

2. USER PROFILE COMPARISON ANALYSIS

To evaluate the system's robustness, I performed cross-profile testing using all six available accounts. This method identifies how the system handles intentional defects in different modules.

User Profile	UI Status	Functional Status	Performance	Final Verdict
standard_user	Perfect	Stable	Fast	PASS
locked_out_user	N/A	Blocked	Fast	PASS (Security)
problem_user	Broken	Unstable	Fast	FAIL (UI)

performance_glitch_user	Perfect	Stable	Very Slow	FAIL (UX)
error_user	Perfect	Critical Failure	Fast	FAIL (Logic)
visual_user	UI Glitches	Stable	Fast	FAIL (Visual)

3. DETAILED BUG REPORTS

BUG-01: Broken Product Images (UI/UX Issue)

- **Summary:** All unique product images are replaced with a generic "broken dog" placeholder.
- **User Profile:** problem_user
- **Steps to Reproduce:** Login with the profile and navigate to the inventory page.
- **Expected Result:** Every product should display its specific high quality image.
- **Actual Result:** All items show the same placeholder image.
- **Severity:** Major

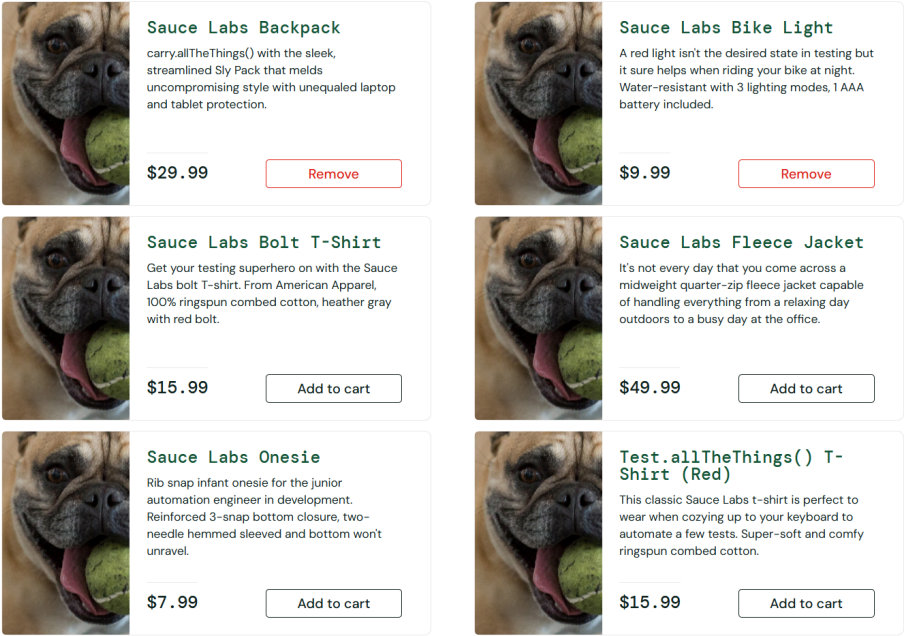


Figure 1: Systemic UI failure where all unique product inventory images are replaced by a generic "broken" placeholder (dog image), preventing visual product verification.

BUG-02: Inconsistent "Add to Cart" and "Remove" Logic

- **Summary:** Some products fail to be added to the cart, and the "Remove" button on the inventory page is non-functional.
- **User Profile:** error_user
- **Steps to Reproduce:** Click on "Add to Cart" for the "Sauce Labs Backpack".
- **Expected Result:** The cart badge should increase to "1".
- **Actual Result:** The click event is ignored, and no item is added.
- **Severity: Critical**

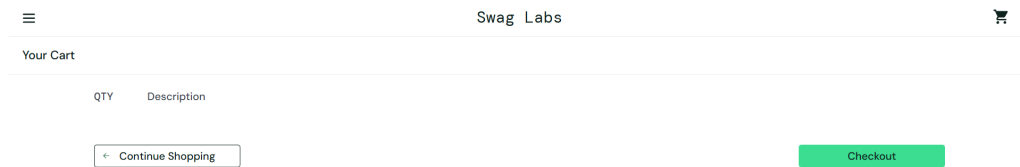


Figure 2: Functional logic defect observed in the cart module; the system fails to sync the inventory state, resulting in an empty cart or nonresponsive "Remove" buttons on the main listing.

BUG-03: Performance Delay (Latency Issue)

- **Summary:** Significant delay (latency) during the login and page transition process.
- **User Profile:** performance_glitch_user
- **Steps to Reproduce:** Enter credentials and click the Login button.
- **Expected Result:** Dashboard should load in under 2 seconds.
- **Actual Result:** Loading takes more than 5 seconds.
- **Severity: Minor/Medium**

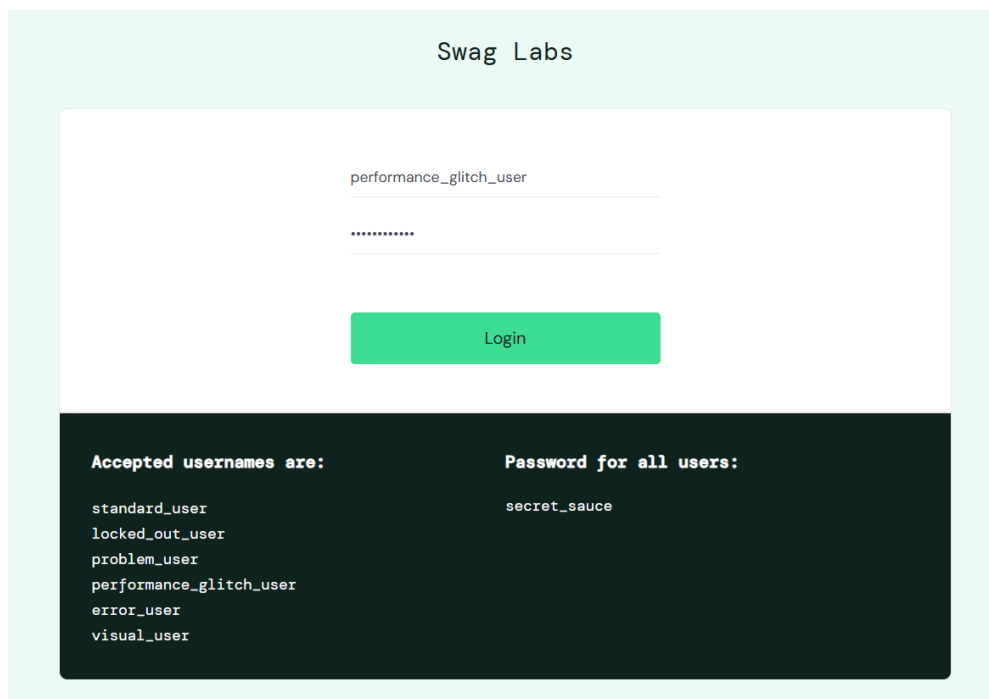


Figure 3: Visual evidence of significant network latency and server side delay during the authentication process, exceeding the 5 second response time threshold.

BUG-04: Visual Regression & Layout Shifts

- **Summary:** Elements on the page are misaligned, causing a poor visual experience.
- **User Profile:** visual_user
- **Steps to Reproduce:** Browse the inventory and checkout pages.
- **Expected Result:** UI elements should be perfectly aligned according to the CSS.
- **Actual Result:** Specific buttons and text fields show layout shifts.
- **Severity:** Minor

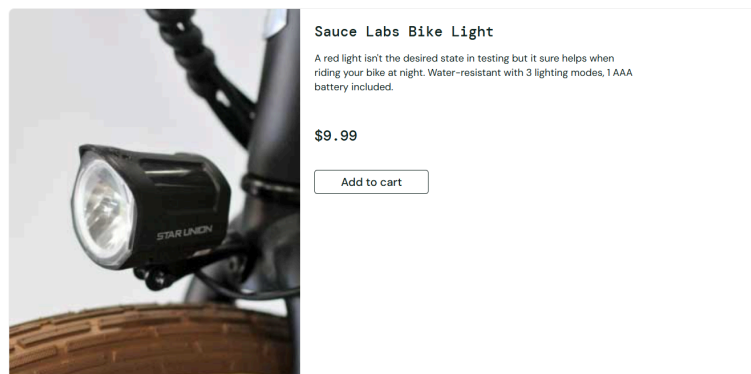


Figure 4: Misalignment of the 'Add to Cart' button and inconsistent margins observed with the visual_user profile, indicating a CSS/Layout regression and poor UI symmetry.

4. SPECIALIZED TESTING RESULTS

Access Control & Security Test

- **Profile:** locked_out_user
- **Objective:** To verify the system's ability to block unauthorized access.
- **Observation:** The system correctly identifies the restricted account and displays the error: "Epic sadface: Sorry, this user has been locked out."
- **Verdict: SUCCESS.** The security policy is functional.

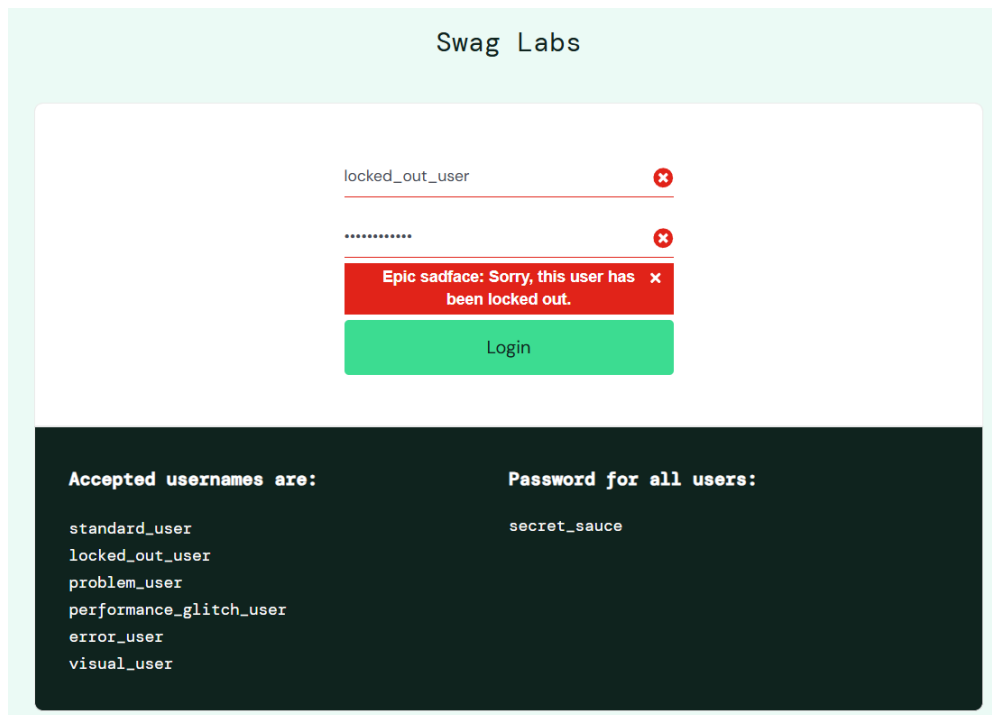


Figure 5: Successful validation of the account lockout policy; the system correctly identifies and blocks restricted credentials

Boundary Value Analysis (Checkout Form)

- **Scenario:** Testing input fields in the Checkout page with numeric values.
- **Observation:** The "First Name" and "Last Name" fields accept numbers (e.g., "12345").
- **Requirement:** These fields should only accept alphabetic characters.
- **Verdict: FAIL (Data Validation Issue).**

The screenshot shows the Swag Labs checkout page. At the top, there is a navigation bar with a hamburger menu icon on the left, the text "Swag Labs" in the center, and a shopping cart icon with a red notification bubble on the right. Below the navigation bar, the text "Checkout: Your Information" is displayed. In the center of the page, there is a form with three input fields. The first field contains the text "Tufan123", the second field contains "99999", and the third field contains "ABCDE". Below the form, there are two buttons: a "Cancel" button with a left arrow icon and a green "Continue" button.

Figure 6a: Entering invalid numeric data into the First Name and Last Name fields on the checkout form to test input validation.

QTY	Description
1	Sauce Labs Bike Light A red light isn't the desired state in testing but it sure helps when riding your bike at night. Water-resistant with 3 lighting modes, 1 AAA battery included. \$9.99

Payment Information:
SauceCard #31337

Shipping Information:
Free Pony Express Delivery!

Price Total
Item total: \$9.99
Tax: \$0.80
Total: \$10.79

[← Cancel](#) [Finish](#)

Figure 6b: Resulting summary page showing that the system accepted the invalid numeric inputs ('Tufan123') without displaying an error message.

5. CONCLUSION

The SauceDemo application is a reliable platform for the standard_user. However, it contains intentional vulnerabilities and defects to test system resilience. The identified Critical functional bugs and Major UI defects demonstrate that the application requires better error handling and visual validation. From a quality assurance perspective, the system meets the basic requirements but fails in edge-case scenarios.